Customer Churn Prediction Analysis

A Machine Learning Approach to Reducing Attrition

Telecom Industry

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1. **Introduction**

Customer churn is the opposite of customer retention. It is when customers totally exit a business, either by canceling subscriptions, not renewing contracts, or ceasing purchases. It is also called **customer attrition**.
Some common reasons customers churn include: **Service quality dissatisfaction, competitive displacement, need elimination, value perception mismatch.**

This report presents a comprehensive machine learning study on customer churn prediction using the WA_Fn-UseC_-Telco-Customer-Churn.csv dataset from Kaggle is used for predicting customer churn in the telecommunications industry, consisting of 7,043 customer records.

The study follows a structured data science workflow: loading of data, exploratory analysis, feature engineering, model training and evaluation, and actionable business recommendations. Four classification algorithms — KNN Classifier, Logistic Regression, Decision Tree and Random Forest — are trained and compared. The best-performing model is analysed in depth to extract both predictive performance and interpretable insights.

1.1 **Research Objectives**

- Develop a binary classification model to predict whether a customer will churn.
- Identify the most influential features driving customer attrition.
- Provide data-driven recommendations for proactive retention strategies
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2. **Data Description**

The dataset used in this project was sourced from kaggle (here) and contains information about customers from a telecommunication company. It consists of 7,043 rows and 21 columns, where each row represents a single customer record.

2.1 Columns Overview

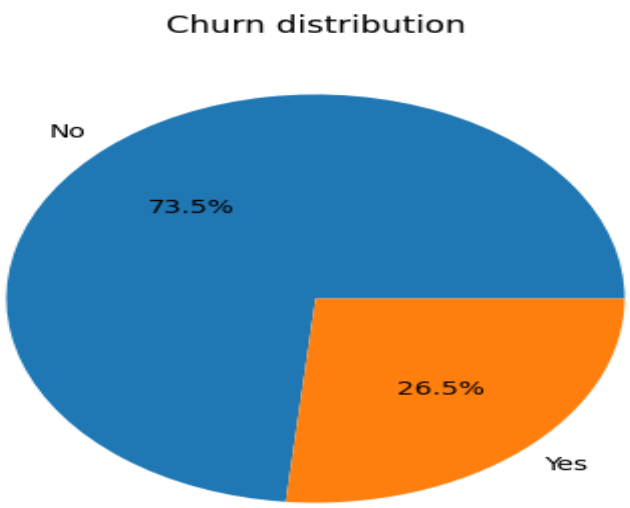
- customerID: Unique identifier assigned to each customer.
- gender: Customer's gender (Male/Female).
- SeniorCitizen: Indicates if the customer is a senior citizen (1 = Yes, 0 = No).
- Partner: Whether the customer has a partner (Yes/No).
- Dependents: Whether the customer has dependents (Yes/No).
- tenure: Number of months the customer has stayed with the company.
- PhoneService: Whether the customer has phone service (Yes/No).
- MultipleLines: Whether the customer has multiple phone lines (Yes/No/No phone service).
- InternetService: Type of internet service (DSL, Fiber optic, No).
- OnlineSecurity: Whether the customer has online security service (Yes/No/No internet service).
- OnlineBackup: Whether the customer has online backup service (Yes/No/No internet service).
- DeviceProtection: Whether the customer has device protection (Yes/No/No internet service).
- TechSupport: Whether the customer has tech support service (Yes/No/No internet service).
- StreamingTV: Whether the customer streams TV (Yes/No/No internet service).
- StreamingMovies: Whether the customer streams movies (Yes/No/No internet service).
- Contract: Type of contract (Month-to-month, One year, Two year).
- PaperlessBilling: Whether the customer uses paperless billing (Yes/No).
- PaymentMethod: Customer's payment method (Electronic check, Mailed check, Bank transfer(automatic), Credit card(automatic))
- MonthlyCharges: Amount charged to the customer monthly.
- TotalCharges: Total amount charged to the customer over time.
- Churn: Indicates whether the customer has left the company (Yes = Churned, No = Retained).

3. **Methodology**

3.1 Exploratory Data Analysis (EDA)
EDA was conducted to understand the structure of the data, uncover patterns, and identify key drivers of churn. The findings below inform subsequent feature engineering and modelling decisions.

3.1.1 Univariate Analysis

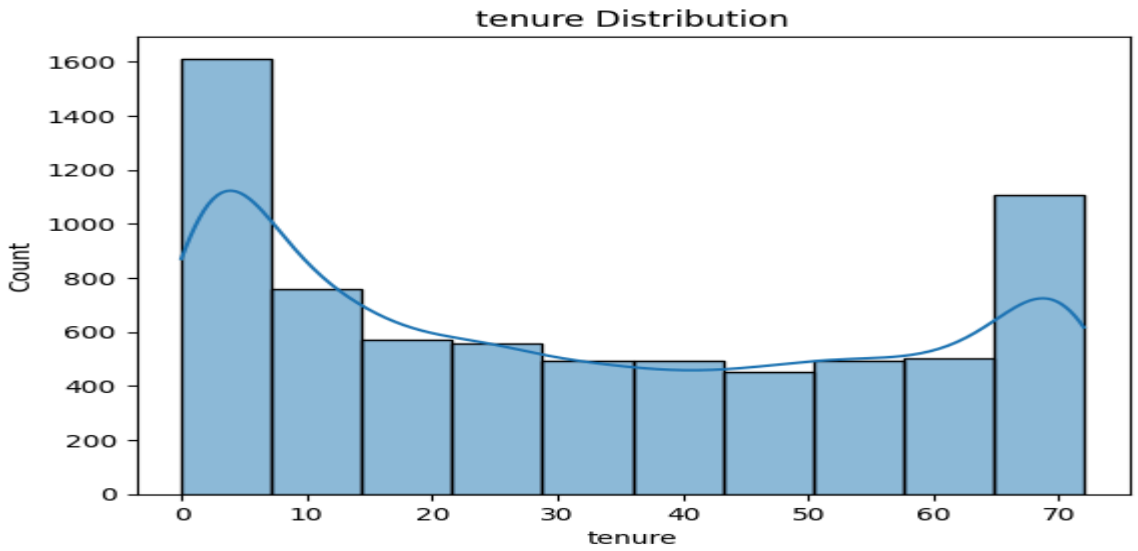
i) Churn distribution
The dataset is imbalanced with a higher proportion of customers who did not churn (at 73.5%) compared to those who churned. This imbalance was considered and handled with a balancing technique as it can affect model performance.



ii) Tenure distribution
The tenure distribution shows a bimodal pattern, with a high concentration of customers in the 0–5 months range and another peak around 65–72 months. This indicates two dominant customer groups:

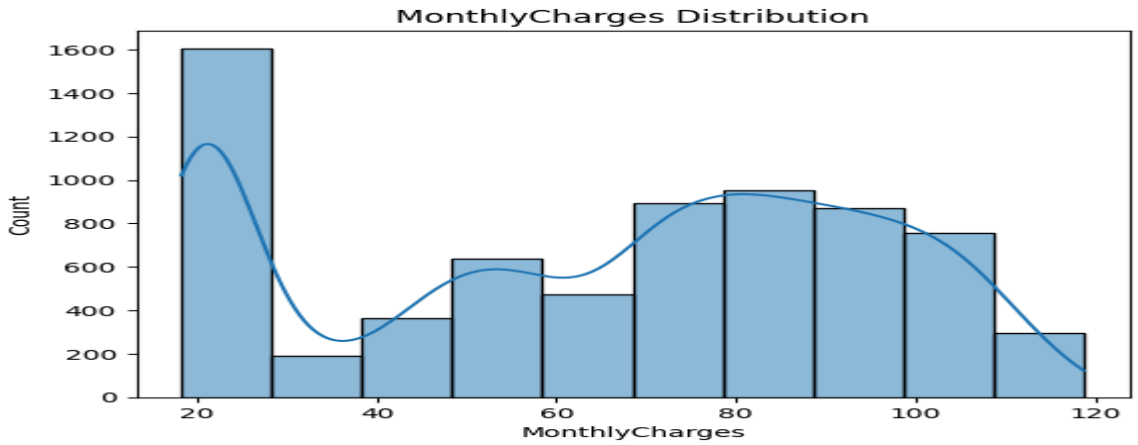
- New customers (low tenure)
- Long-term loyal customers (high tenure)

There is also a noticeable decline in customer count in the mid-tenure range, suggesting that customers are more likely to leave early in their lifecycle before becoming long-term users.



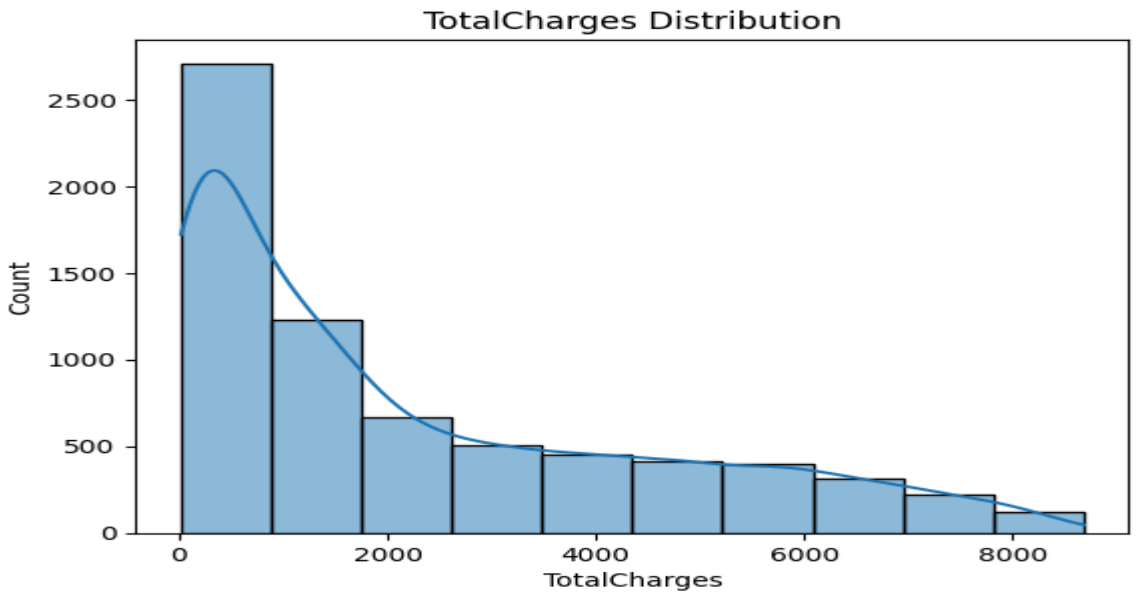
iii) MonthlyCharges distribution

MonthlyCharges distribution shows that prices are not evenly distributed, indicating different pricing ranges. Many customers pay low charges(~20 -30), while another cluster pay high charge(~70 - 100), with a few customers in the middle who pay(~30 -50).This suggests that customers are segmented based on service level which may influence churn behaviour



iv) TotalCharge distribution

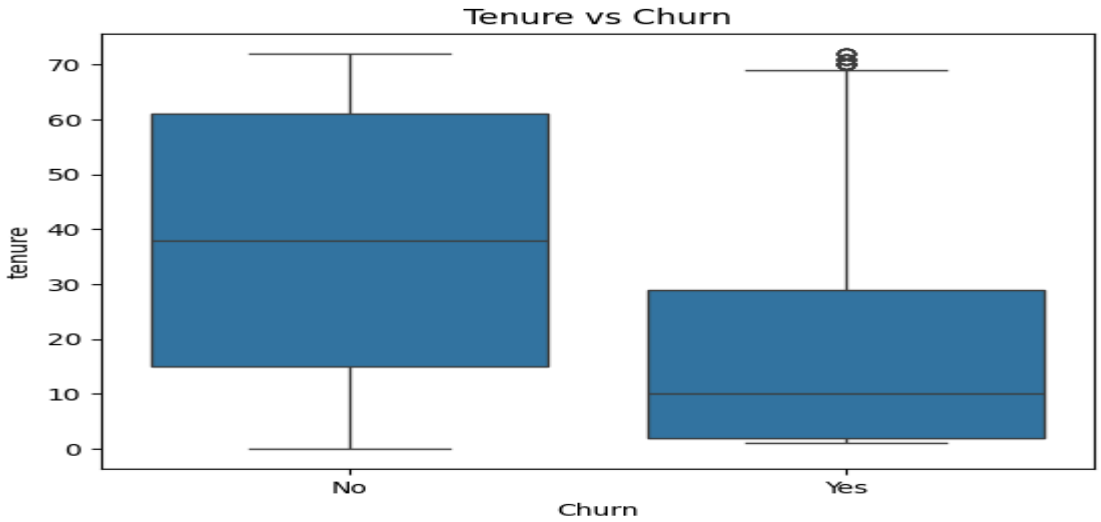
The TotalCharge distribution is right-skewed, with most customers having low cumulative spending. This suggests a large portion of customers are new or have short tenure, while a small portion of long-term customers contribute to higher charges.



3.1.2 Bivariate t Analysis

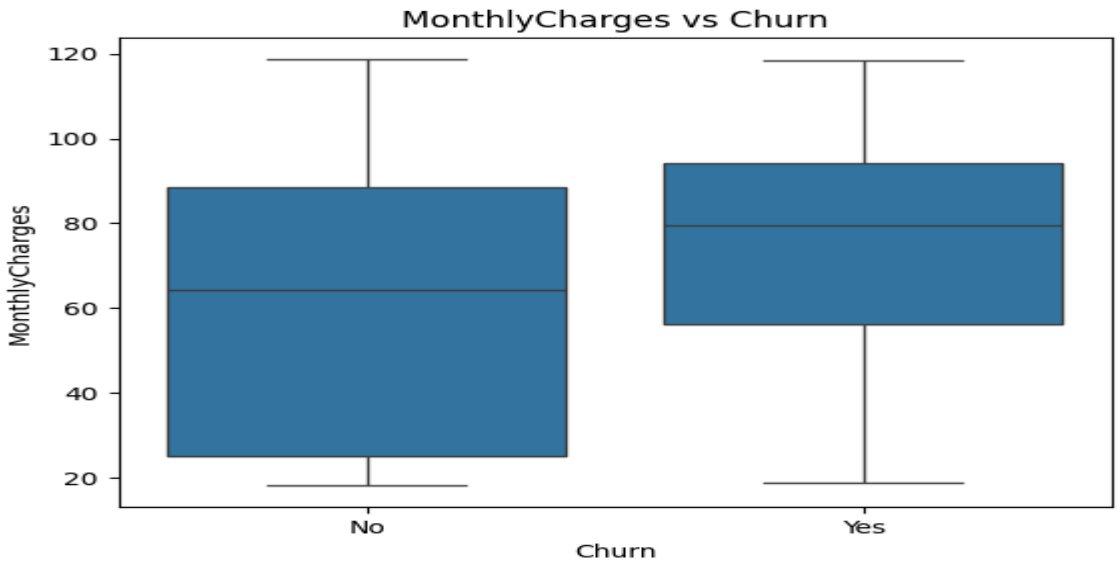
i) Tenure vs Churn

Customers who churn tend to have lower tenure compared to those who remain. This indicates that newer customers are more likely to churn, highlighting the importance of early customer engagement



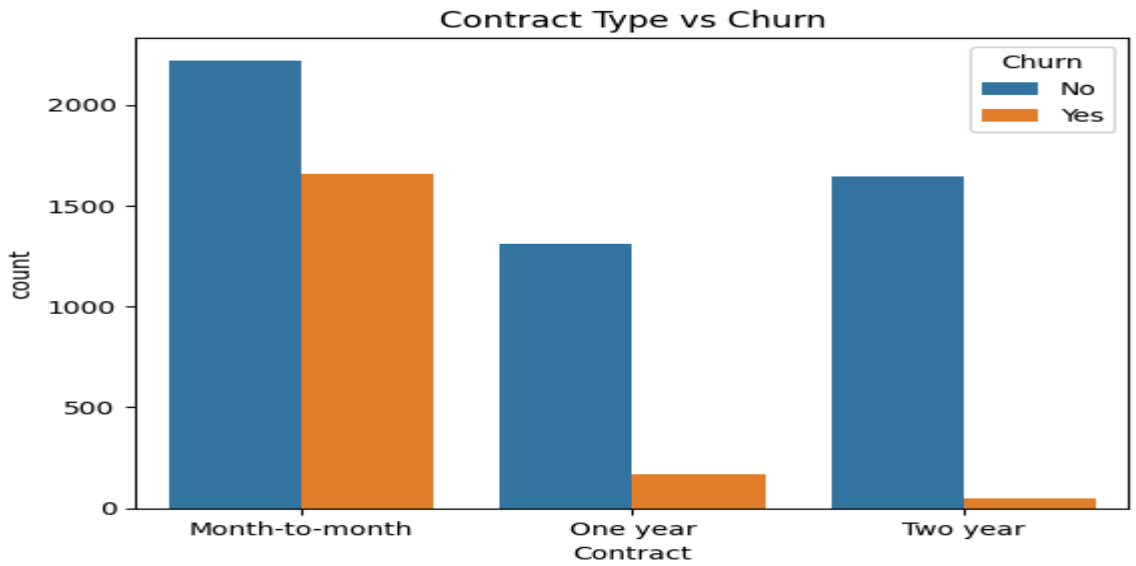
ii) MonthlyCharges vs Churn

Customers with higher monthly charges tend to churn more frequently. This suggests that pricing may play a significant role in customer retention.



iii) Contract Type vs Churn

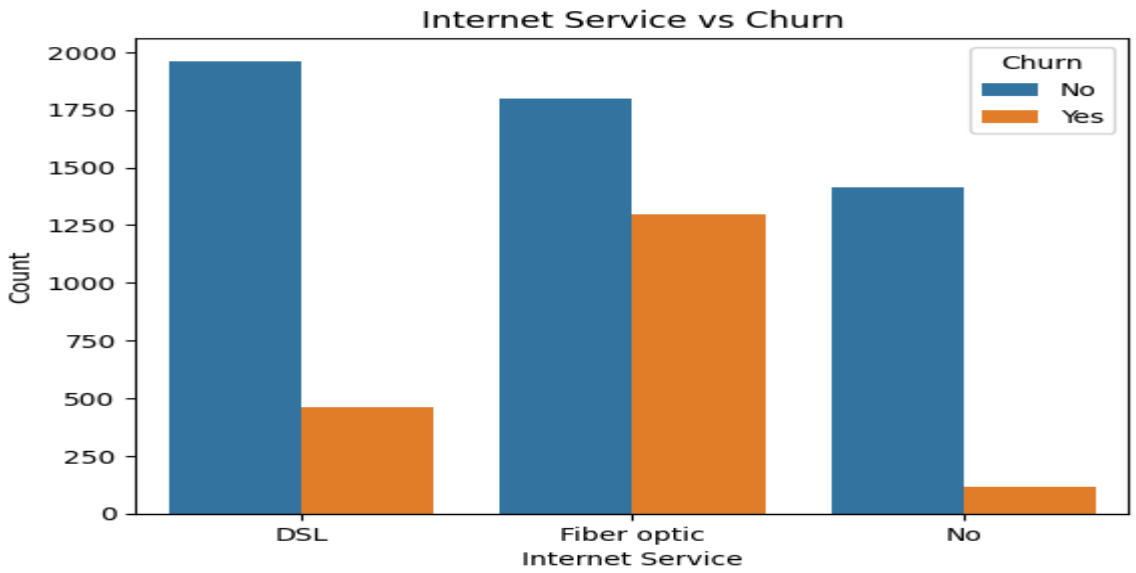
Customers on month-to-month contracts show the highest churn rate, while those on one-year and two-year contracts have significantly lower churn rates. This suggests that long-term contracts improve customer retention, and customers without long term contracts are more susceptible to competitor offers and service dissatisfaction.



iv) Internet Service vs Churn

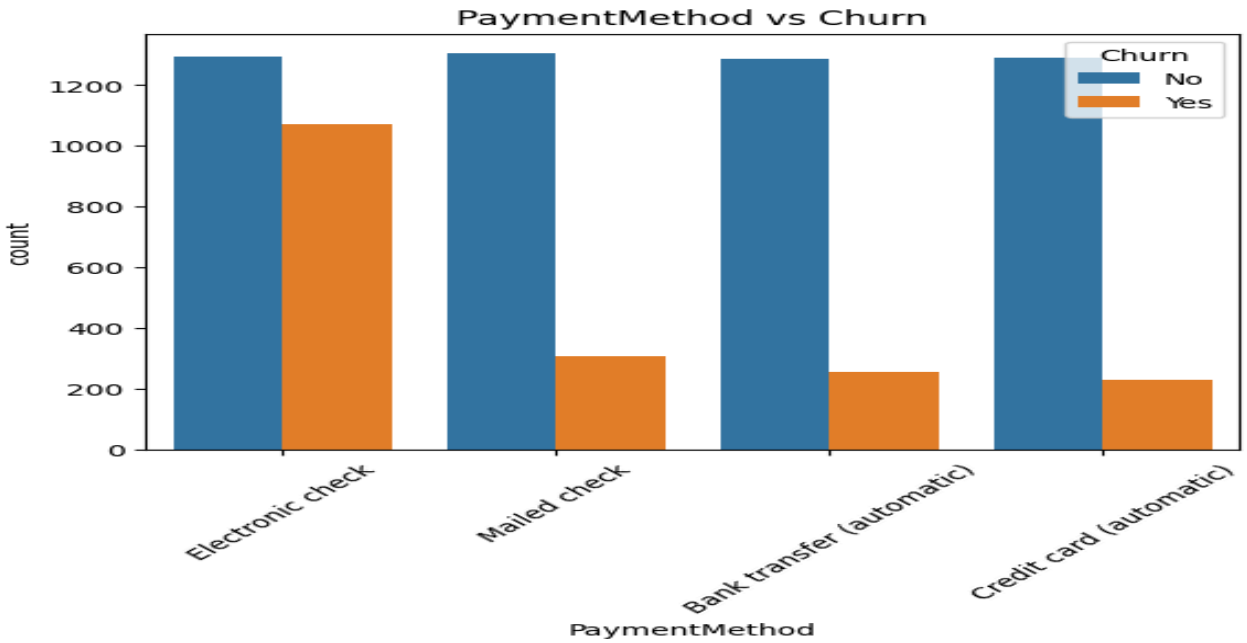
Customers using the Fiber optic plan have twice a higher churn rate compared to DSL users and those without internet service. This may indicate

dissatisfaction with cost, service expectations, or better competitive fibre offerings in the market



v) PaymentMethod vs Churn

Customers using electronic check as their payment method exhibit higher churn rates compared to other payment methods. This may reflect differences in customer engagement or payment convenience.



3.2 Data Cleaning and Preprocessing

- ➢ Missing values: Missing values were observed in the TotalCharges column. Further investigation revealed that these values correspond to customers with zero tenure, indicating that they are new customers who have not yet been billed. Therefore, the missing values were replaced with 0 to reflect the true state of their total charges.
- ➢ The dataset was split into 80% training (5,634 records) and 20% test (1,409 records) sets using stratified sampling to preserve the class ratio in both subsets.
- ➢ One-Hot-Encoding: All categorical variables(except CustomerID which was dropped) were encoded using Panda's dummy variables before being passed to the models.
- ➢ Numerical features (tenure, MonthlyCharges, TotalCharges) were standardised using StandardScaler for KNN Classifier and Logistic Regression, which is sensitive to feature scale. Tree-based models (Decision Tree and Random Forest) received the unscaled features as they are invariant to monotonic transformations.
- ➢ The ***class_weight = 'balanced'*** parameter was used to balance the target in the model. It tells the model to penalise mistakes on the minority class (Yes/churn) more heavily during training. So since "Yes" is ~3x less, errors on "Yes" get ~3x the penalty. This pushes the model to catch more churners, which is usually exactly what you want in a churn problem, as missing a churner is more costly than a false alarm.
- ➢ Feature importances from the Random Forest model were extracted to identify the most influential predictors of churn."

3.3 Models and Evaluation

Four classification algorithms — KNN Classifier, Logistic Regression, Decision Tree and Random Forest were used in this project. GridSearchCV with 5-fold cross-validation (KFold, shuffled), optimising for the F1-score to account for class imbalance, except in Logistic Regression which was trained using default hyperparameters, as the model is relatively robust to tuning on this dataset.

Models were evaluated to determine the best, using metrics such as;

- Accuracy: Proportion of all correctly classified instances.
- Precision: Of all predicted churners, the fraction that actually churned.
- Recall: Of all actual churners, the fraction the model correctly identified
- F1-Score: Harmonic mean of Precision and Recall; balances both concerns.

- AUC-ROC: Area Under the ROC Curve; measures discriminative ability across all thresholds

4 **Results and Key Findings**

4.1 Best Model

	Model	Train Accuracy	Test Accuracy	Precision	Recall	F1 Score	AUC
0	KNN	0.802489	0.772566	0.578035	0.534759	0.555556	0.804902
1	Logistic Regression	0.749867	0.754087	0.524476	0.802139	0.634249	0.840098
2	Decision Tree	0.765333	0.745558	0.514815	0.743316	0.608315	0.820867
3	Random Forest	0.855111	0.769012	0.550725	0.711230	0.620770	0.831931

Four classification algorithms — KNN Classifier, Logistic Regression, Decision Tree and Random Forest were trained on the test set to predict customer churn and were assessed using Accuracy, Precision, Recall, F1-Score, and AUC-ROC, with particular emphasis on **Recall** and **AUC-ROC** as the primary evaluation criteria, since in a churn context, failing to identify a customer who will churn (a false negative) is more costly to the business than incorrectly flagging a loyal customer.

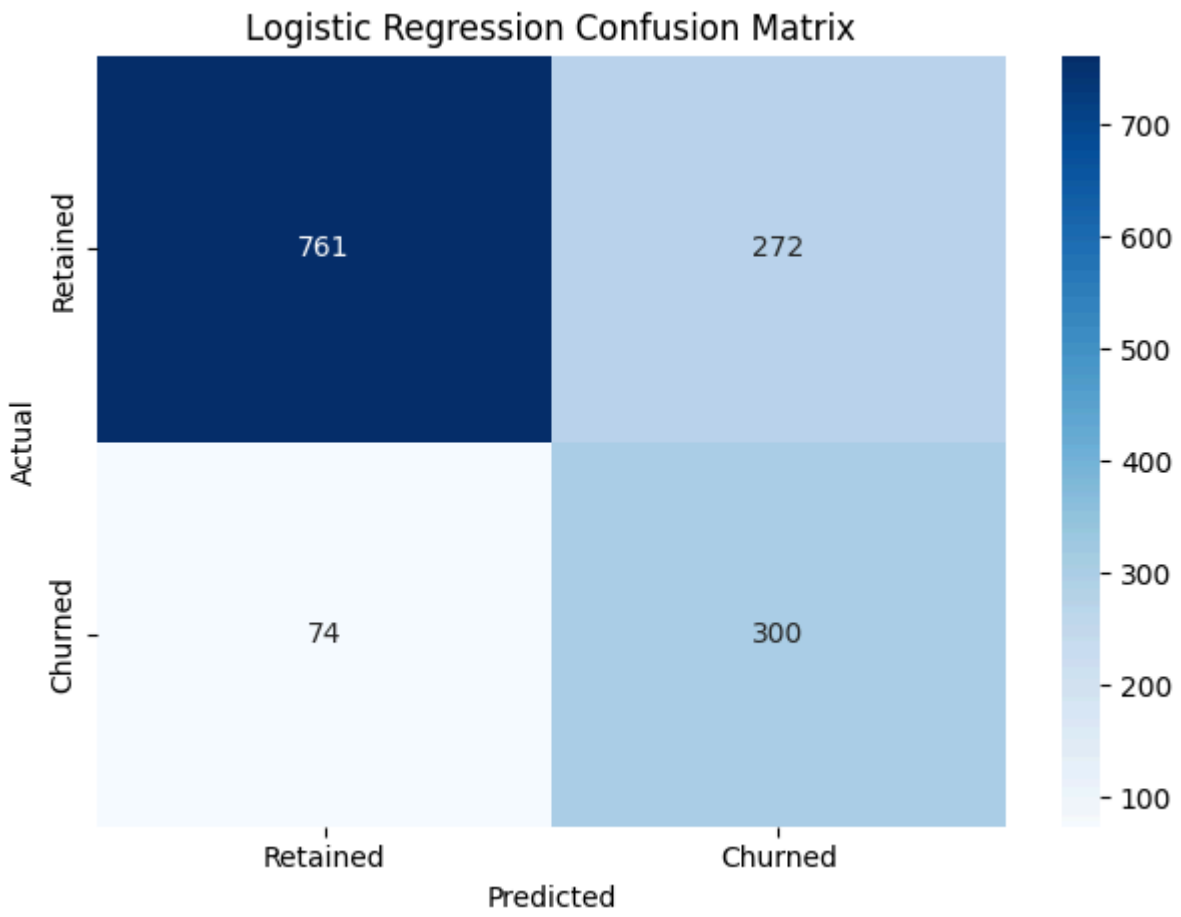
Logistic Regression was selected as the best model, as it achieved the strongest overall performance, achieving the highest AUC-ROC of **0.840** and the highest Recall of **0.802**, meaning it correctly identified approximately 80% of customers who actually churned. Although its precision is comparatively low (0.524) this trade-off is acceptable given the business objective of maximising churn detection. Lastly, its train and test accuracy were also closely aligned (0.750 vs 0.754), indicating that the model generalises well without overfitting.

Random Forest emerged as the second-best model with an AUC-ROC of **0.832**, a Recall of **0.711** and a Precision of **0.551**. However, a larger gap between its training accuracy **(0.855)** and test accuracy **(0.770**) suggests a mild degree of overfitting compared to Logistic Regression.

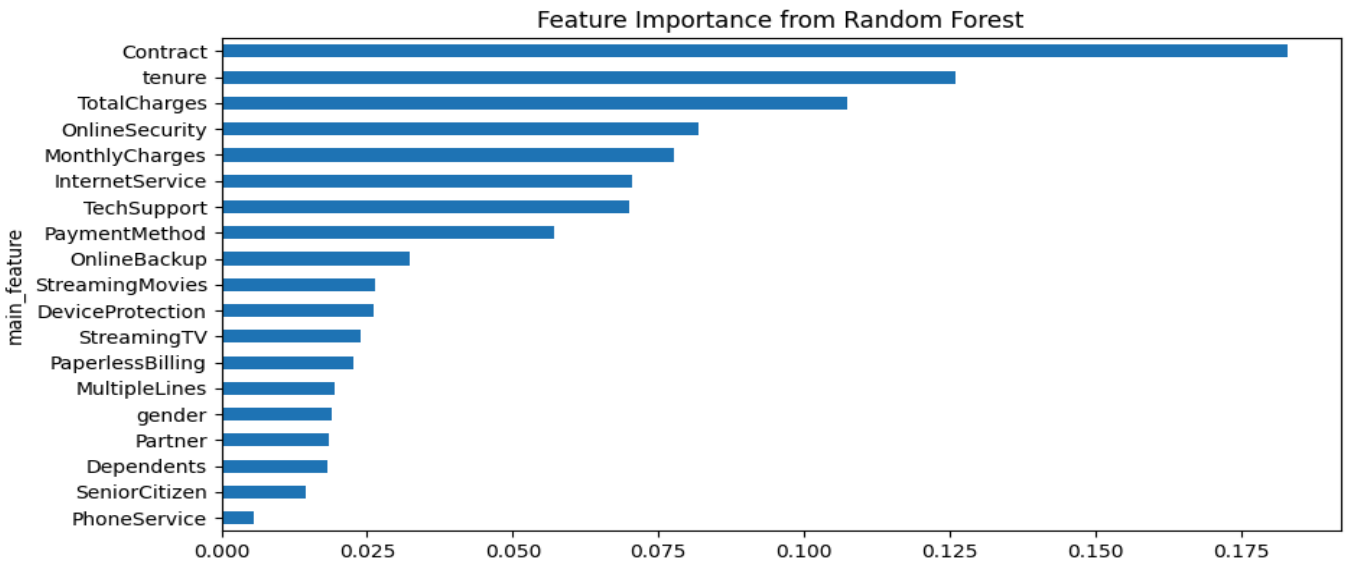
The Decision Tree obtained the lowest test accuracy **(0.769),** a Recall of **0.743** and relatively low Precision **(0.515)**. This reflects a known limitation of single decision trees as they tend to either overfit the training data or, when constrained, generalise poorly. The Random Forest's superior performance over the Decision Tree illustrates the clear advantage of ensemble methods.

The KNN model had a notably high AUC-ROC of **0.805,** but had the least performance among the four models in terms of Recall (0.535) and F1-Score (0.556), indicating it struggled most with identifying true churners.

4.2 Best Model’s Confusion Matrix



4.3 Feature Importance:



The Random Forest ‘Feature Importance’ feature shows that Contract, Tenure and TotalCharges are the most predictive features. This goes in line with the EDA findings: contract type, revenue metric, customer longevity carry the

highest importance, reinforcing the business insight that contractual commitment and service tier are key levers in managing churn.

4.4 Key Finding

- ➢ Customers on month-to-month contracts exhibit the highest churn rate than customers in the other contract types, suggesting that lack of commitment increases likelihood of leaving.

- ➢ Revenue metrics (like TotalCharges and MonthlyCharges) influence churn rate. Customers with higher monthly charges (especially relative to value) are more likely to churn. Indicating potential dissatisfaction with pricing or perceived value.

- ➢ Short-tenure customers are significantly more likely to churn, highlighting the importance of early customer retention.

- ➢ Fiber optic customers are at the highest risk of churners. This possibly due to dissatisfaction with cost, service expectations.

- ➢ Customers paying via electronic check demonstrate higher churn,which may reflect differences in customer engagement or payment convenience.

5 **Conclusion and Recommendation**

5.1 Limitations

While the model shows strong performance, it has some limitations. The dataset captures only one point in time and does not include changes over time, such as seasonal trends or economic conditions. In addition, the model has not been deployed in a real-world setting, so its actual performance may differ. Future work could focus on using more advanced models like XGBoost.

5.2 Recommendation

- Customers on month-to-month contracts should be engaged to switch to long-term plans by offering incentives such as discounts or bonus benefits, as this will help reduce churn.
- Value offered to high-paying customers should be improved by enhancing service quality or providing additional benefits, since customers paying higher charges are more likely to churn if they feel the service is not worth the cost.
- Implementation of strong onboarding programs and early engagement strategies for new customers, as customers with shorter tenure are at a higher risk of leaving.
- Investigation and improvement of fiber optic service experience, pricing and service quality, because customers using this service have a higher likelihood of churning.
- Encourage customers to switch from electronic check to more automated payment methods, such as bank transfers or card payments, to increase customer retention and reduce churn.
- Deploy the trained model in the Customer Relationship (CRM) system to generate weekly churn scores, enabling the retention team to prioritise outreach efficiently.

Reference

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